

OPENRECEIVER

REFERENCE DOCUMENT

Technical sources, documentation, repositories, standards and implementation references for agentic development

Use together with: [OpenReciver.pdf](#)

1. How This Reference Document Must Be Used

This document is the technical reference index for the OpenReceiver agentic development workflow. It is intentionally separate from the product specification. The product specification defines what OpenReceiver must become; this document tells agents where to verify how it should be implemented.

- Do not treat every source as equally authoritative.
- Use primary/official documentation for APIs, standards and platform behavior.
- Use UxPlay and OpenAirPlay as the primary AirPlay implementation/protocol references.
- Use LRCLIB documentation for the automatic lyrics integration, but keep lyrics behind an abstraction.
- Verify licenses before copying, linking or distributing third-party code.
- Never invent undocumented AirPlay behavior. If sources disagree or behavior is unclear, capture evidence and test against a real sender.
- Do not add a dependency simply because a reference document mentions it; use the smallest practical dependency set.

2. Priority Map

Priority	Meaning	Use
P0 — Critical	Agent cannot safely implement the relevant subsystem without it.	Read before implementation.
P1 — Required	Strong implementation/reference source.	Read during the relevant phase.
P2 — Supporting	Useful standards/context/reference.	Consult when needed.
P3 — Future	Only needed for later features.	Do not block v1.

3. AirPlay — Primary Sources

UxPlay — official/current GitHub repository

<https://github.com/FDH2/UxPlay>

Purpose: Primary open-source AirPlay receiver implementation. Study discovery, session negotiation, audio, metadata, artwork, pairing, Windows behavior, GStreamer integration and limitations. | **Priority:** P0 | **Agent:** Protocol / Media / QA

OpenAirPlay — AirPlay specification

<https://openairplay.github.io/airplay-spec/>

Purpose: Protocol reference for publicly documented AirPlay behavior, RTSP flows, audio, metadata, timing, pairing and related structures. | **Priority:** P0 | **Agent:** Protocol

OpenAirPlay — Audio Metadata

<https://openairplay.github.io/airplay-spec/audio/metadata.html>

Purpose: Reference for receiver-side metadata fields such as title, artist, album, artwork and playback-related information. | **Priority:** P0 | **Agent:** Music State / Lyrics

Apple Developer — Developer Downloads

<https://developer.apple.com/download/>

Purpose: Official Apple developer resource; useful when investigating Bonjour/Apple SDK material where appropriate. | **Priority:** P1 | **Agent:** Protocol / Windows

Important AirPlay rule: do not assume Apple Music lyric text is delivered as guaranteed receiver-side AirPlay metadata. The OpenReceiver lyrics subsystem therefore resolves lyrics independently from normalized track identity and playback position.

4. Apple Music / Lyrics References

Apple Support — View lyrics in Apple Music

<https://support.apple.com/en-in/guide/iphone/iphb9bf483aa/ios>

Purpose: Confirms the sender-side Apple Music lyrics experience and provides ecosystem context. Not an OpenReceiver lyrics API. | **Priority:** P1 | **Agent:** Lyrics / Product

Apple Music Provider Support — Lyrics

<https://itunespartner.apple.com/music/support/5225-lyrics>

Purpose: Reference for Apple's synchronized lyrics delivery concepts and timing; research/reference only, not an API dependency. | **Priority:** P1 | **Agent:** Lyrics

LRCLIB

<https://lrclib.net/>

Purpose: Preferred free/keyless automatic lyrics source for the baseline OpenReceiver implementation. | **Priority:** P0 | **Agent:** Lyrics

LRCLIB API/client documentation

<https://lrclib.js.org/>

Purpose: Reference for LRCLIB lookup behavior, parameters, synced lyrics, plain lyrics and client usage. | **Priority:** P0 | **Agent:** Lyrics

LRC format — Wikipedia/reference

[https://en.wikipedia.org/wiki/LRC_\(file_format\)](https://en.wikipedia.org/wiki/LRC_(file_format))

Purpose: Convenient format reference for timestamped lyrics and common tags. Validate parser behavior against real provider output as well. | **Priority:** P1 | **Agent:** Lyrics

5. Networking & Service Discovery Standards

RFC 6762 — Multicast DNS

<https://www.rfc-editor.org/rfc/rfc6762>

Purpose: Standards reference for mDNS, multicast DNS behavior and local discovery. | **Priority:** P1 | **Agent:** Protocol

RFC 6763 — DNS-Based Service Discovery

<https://www.rfc-editor.org/rfc/rfc6763>

Purpose: Standards reference for DNS-SD service advertisement/discovery. | **Priority:** P1 | **Agent:** Protocol

RFC 2326 — RTSP

<https://www.rfc-editor.org/rfc/rfc2326>

Purpose: General RTSP protocol reference. Combine with AirPlay-specific sources rather than treating generic RTSP as the complete AirPlay specification. | **Priority:** P1 | **Agent:** Protocol

RFC 3550 — RTP

<https://www.rfc-editor.org/rfc/rfc3550>

Purpose: RTP packet, timestamp, sequence and synchronization reference for media transport. | **Priority:** P1 | **Agent:** Media

6. Apple Data / Property List References

libplist — official GitHub repository

<https://github.com/libimobiledevice/libplist>

Purpose: Portable C library for Apple property lists, including binary/XML plist handling. Evaluate before adding as a dependency. | **Priority:** P1 | **Agent:** Protocol / Core

libimobiledevice

<https://github.com/libimobiledevice>

Purpose: Related open-source Apple-device protocol ecosystem and supporting libraries; useful for research only unless a concrete dependency is justified. | **Priority:** P2 | **Agent:** Protocol

7. Windows Platform Sources

Microsoft WinUI 3

<https://learn.microsoft.com/windows/apps/winui/>

Purpose: Primary modern Windows UI framework documentation. | **Priority:** P0 | **Agent:** Windows UI

Windows App SDK

<https://learn.microsoft.com/windows/apps/windows-app-sdk/>

Purpose: Application framework, lifecycle, packaging and Windows API integration reference. | **Priority:** P0 | **Agent:** Windows

WASAPI — Windows Audio Session API

<https://learn.microsoft.com/windows/win32/coreaudio/wasapi>

Purpose: Primary Windows audio-output API reference. | **Priority:** P0 | **Agent:** Media / Windows

Microsoft Windows Core Audio

<https://learn.microsoft.com/windows/win32/coreaudio/core-audio-apis>

Purpose: Broader Windows Core Audio reference around endpoints, sessions and audio devices. | **Priority:** P1 | **Agent:** Media

Windows networking / firewall documentation

<https://learn.microsoft.com/windows/security/operating-system-security/network-security/windows-firewall/>

Purpose: Reference for Windows firewall/network behavior that can affect local AirPlay discovery and receiver connectivity. | **Priority:** P1 | **Agent:** Windows / QA

8. GStreamer Sources

GStreamer official documentation

<https://gstreamer.freedesktop.org/documentation/>

Purpose: Primary media pipeline, element, plugin and synchronization reference. | **Priority:** P0 | **Agent:** Media / Windows

GStreamer Application Development Manual

<https://gstreamer.freedesktop.org/documentation/application-development/>

Purpose: Application integration, pipelines, buffers, events, synchronization and playback architecture. | **Priority:** P0 | **Agent:** Media

GStreamer downloads

<https://gstreamer.freedesktop.org/download/>

Purpose: Official binaries/build resources for development and platform setup. | **Priority:** P1 | **Agent:** Windows / Media

GStreamer GitLab

<https://gitlab.freedesktop.org/gstreamer/gstreamer>

Purpose: Source repository for GStreamer itself and implementation/reference work. | **Priority:** P2 | **Agent:** Media

Licensing note: before distributing GStreamer binaries/plugins with OpenReceiver, the Release Agent must verify the applicable licenses and packaging obligations for every included component.

9. Android TV / Android Sources

Android TV developer documentation

<https://developer.android.com/tv>

Purpose: Primary Android TV platform reference. | **Priority:** P0 | **Agent:** Android TV

Compose for TV

<https://developer.android.com/develop/ui/compose/quick-guides/content/create-tv-app>

Purpose: Compose for TV layouts, focus, navigation and TV-specific UI patterns. | **Priority:** P0 | **Agent:** Android UI

Android NDK

<https://developer.android.com/ndk>

Purpose: Native C/C++ development and build integration reference. | **Priority:** P0 | **Agent:** Android / Core

Android JNI tips

<https://developer.android.com/training/articles/perf-jni>

Purpose: JNI boundary, performance and integration guidance. | **Priority:** P0 | **Agent:** Android / Core

Android MediaCodec

<https://developer.android.com/reference/android/media/MediaCodec>

Purpose: Hardware/software codec integration reference for Android media paths. | **Priority:** P1 | **Agent:** Android / Media

Android AudioTrack

<https://developer.android.com/reference/android/media/AudioTrack>

Purpose: Low-level Android PCM/audio output reference when AudioTrack is used. | **Priority:** P1 | **Agent:** Android / Media

Android SurfaceView

<https://developer.android.com/reference/android/view/SurfaceView>

Purpose: Video rendering surface reference for later video support. | **Priority:** P2 | **Agent:** Android / Media

10. Build System / Native Development

CMake documentation

<https://cmake.org/documentation/>

Purpose: Primary build-system reference for the shared C/C++ core and platform integration. | **Priority:** P0 | **Agent:** Core / Build

CMake Getting Started

<https://cmake.org/getting-started/>

Purpose: Practical CMake project/build reference. | **Priority:** P1 | **Agent:** Core / Build

Microsoft Visual Studio C++ documentation

<https://learn.microsoft.com/cpp/>

Purpose: Windows C++ compiler/toolchain reference where MSVC is used. | **Priority:** P1 | **Agent:** Windows / Core

MSYS2

<https://www.msys2.org/>

Purpose: Useful development/build environment reference for studying UxPlay's Windows MinGW workflow; not necessarily the final OpenReceiver toolchain. | **Priority:** P2 | **Agent:** Windows / Protocol

11. Additional Open-Source Protocol / Media References

PJSIP

<https://github.com/pjsip/pjproject>

Purpose: Reference implementation for RTP/STUN/TURN/ICE-related networking. Do not add it unless a concrete OpenReceiver requirement justifies it. | **Priority:** P2 | **Agent:** Protocol / Media

pyatv

<https://github.com/postlund/pyatv>

Purpose: Excellent Apple TV/AirPlay ecosystem research reference, particularly useful for understanding modern Apple-device protocol behavior. Use as research, not automatically as a dependency. | **Priority:** P1 | **Agent:** Protocol

Shairport Sync

<https://github.com/mikebrady/shairport-sync>

Purpose: Open-source AirPlay audio receiver reference, useful for audio-only receiving, synchronization and practical AirPlay behavior. | **Priority:** P1 | **Agent:** Protocol / Media

libav / FFmpeg

<https://ffmpeg.org/>

Purpose: General codec/media reference. Prefer GStreamer/platform APIs where specified by the architecture; do not add FFmpeg solely because it is familiar. | **Priority:** P2 | **Agent:** Media

OpenSSL

<https://www.openssl.org/>

Purpose: Cryptography/TLS library reference if required by a concrete protocol implementation. Follow licensing and platform guidance. | **Priority:** P2 | **Agent:** Core / Security

12. Official GitHub / Source Repositories to Watch

Repository	URL	Why it matters
UxPlay	https://github.com/FDH2/UxPlay	Primary AirPlay receiver reference
OpenAirPlay	https://github.com/openairplay	Open AirPlay protocol/spec ecosystem
libplist	https://github.com/libimobiledevice/libplist	Apple plist parsing
pyatv	https://github.com/postlund/pyatv	Apple TV/AirPlay ecosystem research
Shairport Sync	https://github.com/mikebrady/shairport-sync	AirPlay audio receiver reference
GStreamer	https://gitlab.freedesktop.org/gstreamer/gstreamer	Media framework implementation
PJSIP	https://github.com/pjsip/pjproject	RTP/STUN/TURN/ICE reference

13. Source Authority Rules

- Apple/Android/Microsoft/GStreamer/RFC sources are authoritative for their own APIs/standards.
- OpenAirPlay documents publicly known AirPlay behavior but is not Apple's official protocol specification.
- UxPlay is a practical implementation reference and is especially valuable for interoperability behavior.
- GitHub forks of UxPlay are not authoritative. Prefer the current FDH2/UxPlay repository.
- Community lyric sources are not authoritative for Apple Music's internal lyric timing.
- LRCLIB is a service/provider, not a protocol authority.
- Where documentation and real-device behavior differ, document the discrepancy and reproduce it with a controlled test.
- Licenses must be verified from the current repository/license files before code reuse or redistribution.

14. Recommended Agent Reading Order

1. Read the OpenReceiver Ultimate Final Agentic Development Specification completely.
2. Protocol Agent: UxPlay → OpenAirPlay → mDNS/DNS-SD → RTSP/RTP → pyatv/Shairport Sync as supporting references.
3. Music State Agent: OpenAirPlay metadata → UxPlay metadata behavior.
4. Lyrics Agent: LRCLIB → LRC format → Apple lyrics references.
5. Media Agent: GStreamer → WASAPI → Android MediaCodec/AudioTrack.
6. Windows Agent: WinUI 3 → Windows App SDK → WASAPI → GStreamer.
7. Android Agent: Android TV → Compose for TV → NDK/JNI → MediaCodec/AudioTrack.
8. QA Agent: all relevant protocol + platform docs, then build a real-device test matrix.
9. Release Agent: every dependency's current license, notices and distribution requirements.

15. Minimum Reference Set for v1

If an agent needs the shortest possible set, give it these first:

- UxPlay — <https://github.com/FDH2/UxPlay>
- OpenAirPlay — <https://openairplay.github.io/airplay-spec/>
- OpenAirPlay Audio Metadata — <https://openairplay.github.io/airplay-spec/audio/metadata.html>
- LRCLIB — <https://lrclib.net/>
- LRCLIB docs — <https://lrclib.js.org/>
- GStreamer — <https://gstreamer.freedesktop.org/documentation/>
- WinUI 3 — <https://learn.microsoft.com/windows/apps/winui/>
- WASAPI — <https://learn.microsoft.com/windows/win32/coreaudio/wasapi>
- Android TV — <https://developer.android.com/tv>
- Compose for TV — <https://developer.android.com/develop/ui/compose/quick-guides/content/create-tv-app>
- Android NDK — <https://developer.android.com/ndk>
- RFC 6762 — <https://www.rfc-editor.org/rfc/rfc6762>
- RFC 6763 — <https://www.rfc-editor.org/rfc/rfc6763>
- RFC 2326 — <https://www.rfc-editor.org/rfc/rfc2326>
- RFC 3550 — <https://www.rfc-editor.org/rfc/rfc3550>

16. Important Licensing / Distribution Reminder

This reference document does not grant permission to reuse third-party code. Before incorporating any source into OpenReceiver, the Release Agent must inspect the current license, dependency licenses, linking/distribution conditions and required notices. In particular, UxPlay is GPLv3, so its code cannot simply be copied into an independently licensed architecture without understanding the consequences.

The goal is to learn from and reuse appropriate open-source components while preserving a legally coherent distribution model.